

I. STRATEGY OVERVIEW

Project BWC is a regime-aware quantitative portfolio tracking engine. Built to test structural alpha decay, hierarchical risk parity bisection, and topological model fusion on out-of-sample data. It employs strictly mathematical implementations (DuckDB, Polars, GraphicalLassoCV) to observe algorithmic behavioral bypass and track geometric degradation natively terminating in May 2026.

Report generated: 2026-06-22 05:50 UTC
Initial capital: \$1,000,000 | Benchmark: SPY

II. RISK & RETURN ANALYTICS

Metric	Portfolio	Benchmark (SPY)
Annualized Sharpe	1.31	0.95
Annualized Sortino	3.49	1.30
Calmar Ratio	6.34	1.80
Maximum Drawdown	-9.0%	-22.1%
Cornish-Fisher VaR (5%)	2.56%	-3.88%
Conditional VaR (5%)	-2.01%	-4.50%
Tail Ratio	0.95	0.85
Annualized Return	+68.41%	+10.5%
Annualized Volatility	39.69%	18.0%
Total Return	+7.22%	+22.0%

III. CONFIDENCE-FILTERED SIGNAL ENGINE

The signal engine operates as a monitoring layer that gates signals through a dual-threshold filter:

- Confidence minimum: 0.65
- Fused score minimum: 0.35
- Dynamic accuracy penalty: threshold increases when historical accuracy drops below 70%, making it harder for unreliable tickers to generate actionable signals.

The engine tracks realized vs. expected returns per ticker and uses exponentially-weighted rolling accuracy to ensure only high-quality signals reach the portfolio allocator.

IV. FAMA-FRENCH FACTOR ATTRIBUTION

Factor	Loading	Interpretation
Market (MKT-RF)	1.38	Significant market exposure
Size (SMB)	0.47	Moderate small-cap exposure
Value (HML)	-0.79	Significant value exposure
Momentum (UMD)	-0.17	Mild momentum exposure
FF3 Alpha (ann.)	0.5151	R ² = 0.43
C4 Alpha (ann.)	0.5816	R ² = 0.43

V. PORTFOLIO HOLDINGS

Ticker	Name	Sector	Qty	Avg Cost	Mkt Val	Weight
SPY	SPDR S&P 500 ETF	Broad Market	173	\$674.64	\$129,186.02	13.0%
AMZN	Amazon.com	E-commerce/C	310	\$221.03	\$75,760.90	7.6%
GLD	SPDR Gold Shares	Commodities	450	\$436.80	\$174,204.00	17.5%
GOOGL	Alphabet Inc Class	Big Tech/AI	210	\$316.65	\$77,286.30	7.8%
PLTR	Palantir Technolog	AI Software	438	\$147.10	\$56,269.86	5.7%
XLU	Utilities Select S	Utilities	1200	\$45.74	\$53,712.00	5.4%
XLF	Financial Select S	Financials	700	\$51.25	\$37,499.00	3.8%
XLI	Industrial Select	Industrials	270	\$170.75	\$48,845.70	4.9%
CAT	Caterpillar Inc	Industrials	100	\$766.00	\$98,582.00	9.9%
JPM	JPMorgan Chase	Financials	110	\$307.06	\$35,774.20	3.6%
ESGU	iShares ESG Aware	Broad Market	330	\$146.90	\$53,865.90	5.4%
XOM	Exxon Mobil	Energy	305	\$154.89	\$42,032.05	4.2%
XLE	Energy Select SPDR	Energy	60	\$57.95	\$3,226.20	0.3%
IBB	iShares Biotechnol	Healthcare	120	\$167.12	\$20,836.80	2.1%
NVDA	NVIDIA Corporation	AI Infrastru	274	\$184.04	\$57,729.06	5.8%
ORCL	Oracle Corporation	Software	161	\$155.76	\$29,670.69	3.0%

VI. WALK-FORWARD BACKTEST RESULTS

Metric	naive_equal_wei	topological_hrp	delta
total_return	0.3987	0.3406	N/A
annualized_sharpe	2.7902	3.1870	N/A
annualized_sortino	4.1367	4.7970	N/A
max_drawdown	-0.0536	-0.0502	N/A